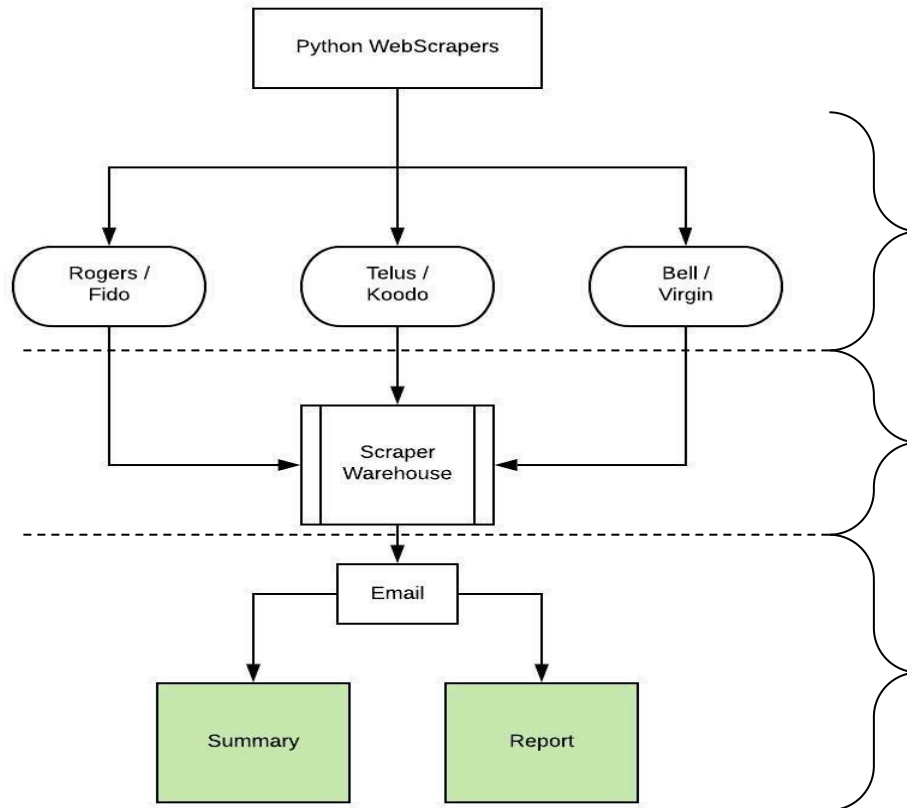


Python Webscraping

Introduction

Objective: To automate the process of collecting competitors pricing information from online sources and to produce a report, along with a summary of changes.

A great amount of manual effort goes into updating competitors pricing information. The task is repetitive, tedious and therefore a perfect candidate for automation. This report describes the process for collecting pricing information through WebScrapper programs and producing a suitable output for management's purposes.



The process flow for the WebScrapper program can be broken down into three phases:

Phase 1:

Individual WebScrapper programs collect pricing information in separate CSV files.

- Individual CSV's are necessary due to differences in webpages. Every webpage is unique and requires a unique approach to scrape.

Phase 2:

CSV files are merged together inside a data-warehouse.

- The data-warehouse may eventually be transformed into a time-series archive to facilitate predictive analysis.

Phase 3:

A report, along with a summary of changes, is produced from the warehouse and distributed via Email.

- The report is formatted for managements standard.
- A summary of changes (attached as a table in the email's body) will isolate and announce any changes from the previous run.

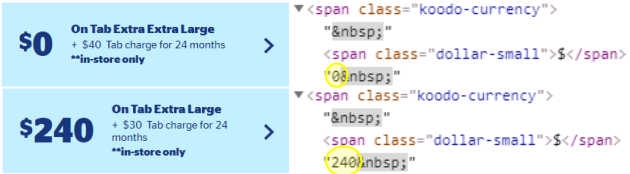
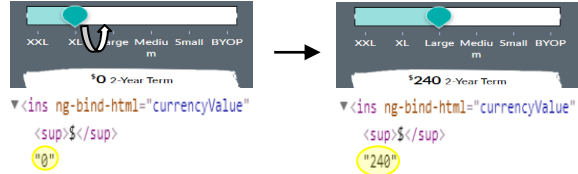
Proposal: To implement WebScrapper programs in conjunction with a data-warehouse and remove all manual effort from the process.

Python Webscraping

Methodology (1)

Phase 1: Individual WebScraper programs collect pricing information in separate CSV files.

Web development has grown rapidly over the past decade, resulting in robust and efficient webpages. AJAX requests are now leveraged by the majority of webpages to populate additional data without an additional server request. Due to the nature of AJAX requests (data not appearing in the source code), advanced functionality must be included in web scraper programs to capture this data.

HTTP 'Regular' Request Webpage	AJAX Request Webpage
- Direct URL (https://www.koodomobile.com/phones/samsung-galaxy-s9) - Quickly loops through URLs for each device - Pricing information is available in source code  - Separate tags for each pricing block	- Open URL (https://www.fido.ca/nac/build-plan) - Slowly clicks individual page elements to navigate to each device - Pricing information collected by pulling a slider bar to create an AJAX request  - The same tag for each slider position

Each WebScraper program outputs a CSV file containing the name of each device and the pricing information for each tier:

Device	XXL	XL	L	M	S
Samsung Galaxy A70 128GB	\$0	\$0	\$0	\$0	\$120
iPhone 7 32GB	\$0	\$0	\$0	\$0	\$120
Huawei P30 lite 128GB			\$0	\$0	\$0
Samsung Galaxy S9 64GB	\$0	\$0	\$0	\$0	\$120
iPhone 11 64GB	\$0	\$280	\$479	\$640	\$760

Capabilities for each individual WebScraper program are shown below:

Brand Website	Processing Time (mins)	# of Devices	Build Time (days)
Rogers	8.66	29	6
Fido	10	28	6
Telus	7.15	36	6
Koodo	2	36	2
Bell	0.30	40	1
Virgin	3	37	1

Python Webscraping

Methodology (1)

Phase 2: CSV files are merged together inside a data-warehouse.

The data-warehouse has a 'short-term memory'. As new input is fed in and the warehouse is updated, the input is compared to that of the previous run. If there is a price change (a difference in value between the two runs), it is captured in the summary of changes. The potential for 'long-term memory' is in building towards a time-series archive.

Bell	SMART PAY			
Device	Bell	Telus	Rogers	Gaps / Leads
Apple iPhone 11 Pro Max 64GB	\$64.80	64.75	64.79	-0.05
Apple iPhone 11 Pro 64GB	\$58.96	58.92	58.96	-0.04
Apple iPhone 11 64GB	\$41.88	41.79	41.88	-0.09
Apple iPhone XS 64GB	\$25.00	25.00	53.54	0.00
Apple iPhone XS Max 64GB	\$29.00	40.00	59.38	11.00
Apple iPhone XR 64GB	\$34.17	34.17	33.75	-0.42
Apple iPhone 8 64GB	\$25.63	25.62	25.41	-0.22
Apple iPhone 7 32GB	\$15.00	15.00	20.41	0.00
Apple iPhone 6S 32GB	\$10.00	10.00	N/A	0.00
Samsung Galaxy Note10+ 256GB	\$60.83	54.58	71.88	-6.25
Samsung Galaxy S10+ 128GB	\$51.25	50.83	65.63	-0.42
Samsung Galaxy S10e 128GB	\$35.42	35.42	46.88	0.00
Samsung A70	\$15.00	24.58	29.17	9.58
Samsung A50	\$17.50	15.00	20.83	-2.50
Samsung A20	\$13.33	13.96	16.67	0.63

Virgin	Gold			
Device	Virgin	Fido	Koodo	Gold Gap
iPhone 7 32GB	0	0	0	0
iPhone 8 64GB	240	250	240	0
iPhone XR 64GB	430	450	425	5
iPhone Xs 64GB	300	220	600	80
iPhone Xs Max 64GB	320	650	750	-330
iPhone 11 64GB	600	640	595	5
iPhone 11 Pro 64GB	980	1050	975	5
iPhone 11 Pro Max 64GB	1110	1190	1105	5
Samsung Galaxy A20 32GB	0	0	0	0
Samsung Galaxy A50 64GB	0	0	0	0
Samsung Galaxy A70 128GB	0	0	120	0
Samsung Galaxy S9 64GB	120	120	120	0
Samsung Galaxy S10 128GB	700	720	635	65
Samsung Galaxy S10+ 128GB	800	870	960	-70
Samsung Galaxy S10e 128GB	120	120	360	0

Phase 3: A report, is produced from the warehouse and distributed via Email.

The final output is distributed via email. The output contains both the formatted report (containing all pricing) and the summary of changes in the body of the email. If no updates are encountered, the report will still send.

The WebScraper program may run regularly at any given time, given that a computer is left on and connected to the internet so that it may run the script. Windows Task Scheduler has already been setup and tested to preform the process without any human interaction.

Fri 1/17/2020 10:15 AM

Ro Boto <roboto1324876@outlook.com>

[EXT]Price changes have occurred!

To: Ro Boto; Chiappori, Daniel

If there are problems with how this message is displayed, click here to view it in a web browser.

FlankerReport.xlsx
7 KB

Dearest colleagues,

The following price changes have occurred:

Unnamed: 0	Brand	Device	Plan	Old_Price	New_Price	
0	0	Fido	Samsung Galaxy A70 128GB	Medium	\$60	\$0
1	1	Fido	Samsung Galaxy A70 128GB	Small	\$180	\$120
2	2	Fido	Samsung Galaxy S9 64GB	Medium	\$120	\$0
3	3	Fido	Samsung Galaxy S9 64GB	Small	\$240	\$120

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Features & Add-Ons

The following chart summarizes the base features of the WebScraper:

Feature	Description
Data Warehouse (Short-Term Memory)	- Collective pricing information is stored in it's own space after each run. - Facilitates the 'Summary of Changes' feature.
Summary of Changes	- As the WebScraper collects new pricing information, it will check the previous run's values to check for any differences (ie. Price changes). - If a price change is identified, it will be isolated and 'called out' in the body of the email output.
Formatted Report	An aesthetic report which includes conditional filters to highlight gaps and leads between competitors.
Email Bot	An automated email which includes the summary and report.

Supplementary features (Add-ons) may be incorporated into the program if desired. The below table lists several potential Add-ons, along with an estimated procedure to implement and time to completion:

Add-On	Description	Steps Involved	Time To Completion
TCO Output	Total cost of ownership report.	- Creating a report to effectively display TCO. - Leveraging warehouse data and feeding it in.	1 Month
Time-Series Warehouse (Long-Term Memory)	Archiving pricing information in a time-series.	- Updating the data-warehouses infrastructure. - New logic for summary of changes to avoid duplication.	2 Months
Different Regions	Regional pricing (ie. Quebec)	Expanding on WebScraper logic.	1 Month
Additional Memory Sizes	Pricing for all memory sizes for each device.	Expanding on WebScraper logic.	1 Month
'CronJob' Trigger Update	The ability to respond with a keyword to the email bot and have it run the program for you.	- Investigating alternatives to Task Scheduler. - Investigating querying Outlook inbox to look for trigger word.	4 Months